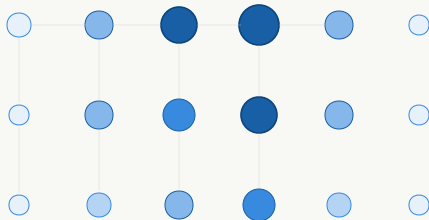


Discrete lattice

$$\text{hopping } T = \hbar^2 / (4m\Delta s^2)$$



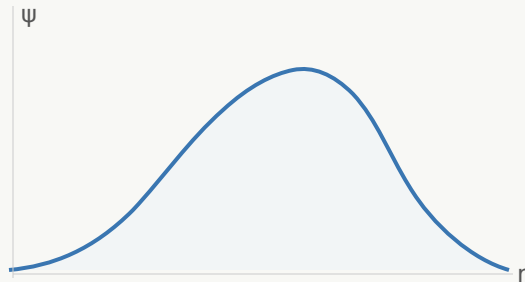
$$\begin{aligned} \psi_i(t+\Delta t) &= \psi_i(t) \\ &- (i\Delta t/\hbar) H_{\text{lattice}} \psi \end{aligned}$$

complex hopping

$\Delta s \rightarrow 0$
→
continuum

Continuum limit

$$\nabla^2 = \text{lim of lattice Laplacian}$$



$$\begin{aligned} i\hbar \partial\psi/\partial t &= H\psi \\ H &= -\hbar^2\nabla^2/(2m) + V \end{aligned}$$

Schrödinger equation